## *Summary*

- A virtual machine fails if the underlying hardware or virtualization layer fails.
- You can recover a failed virtual machine with the help of a CloudWatch alarm: by default, data stored on EBS, as well as the private and public IP addresses, stays the same.
- An AWS region consists of multiple isolated groups of data centers called availability zones.
- Recovering from a data center outage is possible when using multiple availability zones.
- Use autoscaling to replace a failed virtual machine, even in the event of a data center outage. The pitfalls are that you can no longer blindly rely on EBS volumes and, by default, IP addresses will change.
- Recovering data in another availability zone is tricky when stored on EBS volumes instead of managed storage services like RDS, EFS, S3, and DynamoDB.
- Some AWS services use multiple availability zones by default, but virtual machines run in a single availability zone.